

Faithfulness and Preprocessing Robustness of Spectrogram Attributions in Bioacoustic Classification

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Abstract

Keywords: ecoacoustics, explainable artificial intelligence, bioacoustic monitoring, spectrogram attribution, preprocessing robustness, ecological machine learning

1. Introduction

2. Related work and conceptual background

- 2.1. Deep learning and ecoacoustic monitoring*
- 2.2. Explainability for spectrotemporal ecological data*
- 2.3. Faithfulness, stability, and preprocessing dependence*

3. Materials and methods

- 3.1. Study system and acoustic dataset*
- 3.2. Acoustic representations and preprocessing regimes*
- 3.3. Classification model and evaluation protocol*
- 3.4. Explanation methods*
- 3.5. Faithfulness and robustness evaluation*
- 3.6. Statistical analysis*
- 3.7. Reproducibility and data/code availability*

4. Results

- 4.1. Predictive performance across preprocessing regimes*
- 4.2. Attribution stability across representations*
- 4.3. Faithfulness of explanations*
- 4.4. Ecologically relevant spectrotemporal patterns*
- 4.5. Sensitivity and robustness analyses*

5. Discussion

- 5.1. What preprocessing dependence means for ecological inference*
- 5.2. When stable predictions do not imply stable explanations*
- 5.3. Implications for explainable ecoacoustic monitoring*
- 5.4. Methodological recommendations*
- 5.5. Limitations and external validity*

6. Conclusions

CRedit authorship contribution statement

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

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References